

Pranav Guhathakurta

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EDUCATION

University of Pennsylvania – School of Engineering and Applied Science <i>M.S.E. in Computer Science</i>	Philadelphia, PA <i>Aug. 2026 – Dec 2027</i>
University of Pennsylvania – The Wharton School <i>B.S. in Economics – Finance; Minor in Computer Science</i>	<i>Aug. 2022 – May 2026</i>
• GPA: 3.51/4.00; Joseph Wharton Scholar; Benjamin Franklin Scholar • Coursework: Data Structures & Algorithms, Computer Systems, Programming Languages, Mathematical Foundations of CS, Databases, Statistics, Quantitative Modeling	

EXPERIENCE

Flagright (Y Combinator S22) <i>Software Engineer Intern</i>	San Francisco, CA <i>May 2024 – Aug. 2024</i>
• Engineered a Node.js microservice on AWS API Gateway for real-time sanctions screening, processing 25K+ daily transactions with <800 ms latency • Deployed financial crime screening services on-premise for a sovereign fund, integrating application workflows with secure local infrastructure • Supported production readiness, reliability, and compliance across 5+ services through SOC 2 audit preparation and business continuity planning	
Iridium Business Solutions <i>Software Engineer Intern</i>	Cape Town, South Africa <i>May 2023 – Jul. 2023</i>
• Developed a multi-tenant financial analytics platform using React, AWS Lambda, S3, and DynamoDB, supporting 50+ corporate entities • Built 10+ REST API endpoints and implemented AWS Cognito authentication and authorization for 1K+ daily requests • Optimized cloud-based backend request processing, reducing response time by 30%	
Joseph Wharton Scholars Research <i>Quantitative Researcher</i>	Philadelphia, PA <i>Aug. 2025 – Present</i>
• Developed a Python-based financial transaction screening system using Node.js and Neo4j for relationship-based risk detection across financial entities • Built reproducible data ingestion, preprocessing, and retrieval pipelines across heterogeneous financial datasets • Implemented and evaluated semantic entity resolution using embeddings and cosine similarity, achieving 92% recall	

PROJECTS

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TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL
AI & Machine Learning: LLM APIs, RAG, Prompt Engineering, Embeddings, Vector Search, Scikit-learn, Model Evaluation
Backend & Data: Node.js, FastAPI, REST APIs, Data Pipelines, PostgreSQL, MongoDB, DynamoDB, Neo4j
Cloud & Infrastructure: AWS, Docker, Git

ACTIVITIES & INTERESTS

Technology & Finance: NeurIPS Attendee FinTech Global Event Assistant Quantitative Finance Financial Markets
